

5. Start the Glue Job

The `boto3` Glue client is used to start a Glue job with specific arguments:

```
python

glue = boto3.client('glue')
job_name = 'csvdata'
arguments = {
    '--s3_path': s3_path,
    '--buck': s3_bucket,
    '--obj': s3_key
}

response = glue.start_job_run(JobName=job_name, Arguments=arguments)
```

- `boto3.client('glue')` : Creates a Glue client.
- `job_name` : Specifies the Glue job to be triggered (replace `'csvdata'` with the actual Glue job name).
- `arguments` : Provides key-value pairs as arguments to the Glue job:
 - `--s3_path` : Full S3 path to the uploaded file.
 - `--buck` : The name of the S3 bucket.
 - `--obj` : The S3 object key (file name).
- `glue.start_job_run()` : Starts the Glue job with the specified job name and arguments.

Create lambda:

Start Glue job ,JOBNAME---myjobsma:

Whenever there is a new s3 file loaded into bucket, lambda is triggered

```
lambda_function.py
1  import json
2  import boto3
3
4  def lambda_handler(event, context):
5      try:
6          # Get the S3 bucket and key from the event
7          s3_bucket = event['Records'][0]['s3']['bucket']['name']
8          s3_key = event['Records'][0]['s3']['object']['key']
9          s3_path = f"s3://{s3_bucket}/{s3_key}"
10         print('glue sendimng path : ', s3_path)
11         print('glue sendimng bucket : ', s3_bucket)
12         print('glue sendimng key : ', s3_key)
13
14         # Staadfasdftr the Glue job
15         glue = boto3.client('glue')
16         job_name = 'myjobsma'
17         arguments = {
18             '--s3_path': s3_path,
19             '--buck': s3_bucket,
20             '--obj': s3_key
21         }
22         response = glue.start_job_run(JobName=job_name, Arguments=arguments)
23         Amazon Q Tip 2/3: Invoke suggestions with [Option] + [C] ([ESC] to exit)
24     except KeyError as e:
25         print("Error: Missing key in event payload:", e)
26         # Handle the error gracefully, e.g., by logging or returning an error response
```

-- cloudwatch logs ,
showing s3 file details details from lambda cloudwatch logs

No older events at this moment. Retry	
2025-01-22T16:02:27.993Z	INIT_START Runtime Version: python:3.12.v38 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:7515e00d6763496e7a147ffa395e
2025-01-22T16:02:28.315Z	START RequestId: 27c1bcfd-e27b-4f38-9243-83157fbd8682 Version: \$LATEST
2025-01-22T16:02:28.316Z	glue sendimng path : s3://mithelesh2023/livedata/uber.csv
2025-01-22T16:02:28.316Z	glue sendimng bucket : mithelesh2023
2025-01-22T16:02:28.316Z	glue sendimng key : livedata/uber.csv
2025-01-22T16:02:31.339Z	END RequestId: 27c1bcfd-e27b-4f38-9243-83157fbd8682
2025-01-22T16:02:31.339Z	REPORT RequestId: 27c1bcfd-e27b-4f38-9243-83157fbd8682 Duration: 3000.00 ms Billed Duration: 3000 ms Memory Size: 128 MB Max Me
No newer events at this moment. <i>Auto retry paused.</i> Resume	

Create role:

ir

Configuration

Aliases

Versions

Execution role

Role name

lambdaccesss3

Edit

View

Resource summary

To view the resources and actions that your function has permission to access, choose a service.

AWS Application Auto Scaling

1 action, 1 resource

By action

By resource

Resource	Actions
All resources	Allow: application-autoscaling:DescribeScalingPolicies

Role: attach this role in lambda ,under permissions

lambdaaccess3 [Info](#)

Allows Lambda functions to call AWS services on your behalf.

Summary

Creation date

March 08, 2024, 20:05 (UTC-05:00)

Last activity

37 minutes ago

ARN

arn:aws:iam::123456789012:role/lambdaaccess3

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (7) [Info](#)

You can attach up to 10 managed policies.



[Simulate](#)

[Remove](#)

Search		Filter by Type	
		All types	
☐	Policy name	Type	Attached entities
☐	AmazonEC2FullAccess	AWS managed	5
☐	AmazonS3FullAccess	AWS managed	10
☐	AWSGlueServiceRole	AWS managed	1
☐	AWSLambda_FullAccess	AWS managed	2
☐	CloudWatchEventsFullAccess	AWS managed	1
☐	CloudWatchFullAccess	AWS managed	2
☐	CloudWatchFullAccessV2	AWS managed	2

► **Permissions boundary** (not set)

Create AWS Glue job:

This glue job `job_name = 'myjobsma'` is passed from lambda.

[Script](#) | [Job details](#) | [Runs](#) | [Data quality](#) | [Schedules](#) | [Version Control](#) | [Upgrade analysis -](#)

Script Info

```
1 import sys
2 from awsglue.transforms import *
3 from awsglue.utils import getResolvedOptions
4 from pyspark.context import SparkContext
5 from awsglue.context import GlueContext
6 from awsglue.job import Job
7 from pyspark.sql.functions import *
8 ## @params: [JOB_NAME]
9 args = getResolvedOptions(sys.argv, ['s3_path','buck','obj'])
10 s3_path= args['s3_path']
11 bucket = args['buck']
12 file_path = args['obj']
13 import os
14
15 allpath = os.path.basename(file_path)
16
17 # Split the filename and extension
18 filename, extension = os.path.splitext(allpath)
19 tab=filename
20 print('tablename:',tab)
21
22 sc = SparkContext()
23 glueContext = GlueContext(sc)
24 spark = glueContext.spark_session
25 #data="s3://dfdfsd"
26
27 df=spark.read.format("csv").option("header","true").option("inferSchema","true").load(s3_path)
28 df.show()
```

Python | Ln 1, Col 1 | Errors: 0 | Warnings: 0

AWS Glue job IAM role:
Create role and add in details under glue job

mygluecrawler [Info](#)

Allows Glue to call AWS services on your behalf.

Summary

Creation date
March 09, 2024, 20:03 (UTC-05:00)

Last activity
 11 minutes ago

ARN
 arn:aws:iam::123456789012:role/mygluecrawler

Maximum session duration
1 hour

- Permissions
- Trust relationships
- Tags
- Last Accessed
- Revoke sessions

Permissions policies (9) [Info](#)

You can attach up to 10 managed policies.

Q Search

Filter by Type
All types

☐	Policy name	Type	Attached entities
☐	AmazonRedshiftFullAccess	AWS managed	1
☐	AmazonS3FullAccess	AWS managed	10
☐	AWSGlueConsoleFullAccess	AWS managed	1
☐	AwsGlueDataBrewFullAccessPolicy	AWS managed	1
☐	AWSLambda_FullAccess	AWS managed	2
☐	AWSLambdaBasicExecutionRole	AWS managed	1
☐	AWSLambdaRole	AWS managed	1
☐	CloudWatchFullAccess	AWS managed	2
☐	CloudWatchFullAccessV2	AWS managed	2

Pass role in glue job details

Script

Job details

Runs

Data quality

Schedules

Version Control

Upgrade analysis - preview

Basic properties

Info

Name

myjobsma

Description - optional

Descriptions can be up to 2048 characters long.

IAM Role

Role assumed by the job with permission to access your data stores. Ensure that this role has permission to your Amazon S3 sources, targets, temporary directory, scripts, and any libraries used by the job.

mygluecrawler

Type

The type of ETL job. This is set automatically based on the types of data sources you have selected.

Spark

Glue version

Info

Glue 4.0 - Supports spark 3.3, Scala 2, Python 3

```
args = getResolvedOptions(sys.argv, ['s3_path', 'buck', 'obj'])
s3_path= args['s3_path']
bucket = args['buck']
file_path = args['obj']
```

 Copy  Edit

- `getResolvedOptions()` : Retrieves the command-line arguments passed to the Glue job.
 - `s3_path` : Full S3 path of the file.
 - `buck` : Name of the S3 bucket.
 - `obj` : File name or object key.
-

Extracting the File Name

```
python

allpath = os.path.basename(file_path)
filename, extension = os.path.splitext(allpath)
tab = filename
print('tablename:', tab)
```

 Copy  Edit

- `os.path.basename()` : Extracts the file name from the full path.
- `os.path.splitext()` : Splits the file name into the base name (`filename`) and the extension (`extension`).
- `tab` : Represents the table name, inferred from the file name (without the extension).

Cloudwatch logs

  EC2  RDS  S3  Lambda  EMR

CloudWatch > Log groups > /aws-glue/jobs/output > jr_4c62c51fda41ee7f7c5714fa2f038b3ec1c07e108c32032dc7a5575e587d4353

CloudWatch <

favorites and recents ▶

dashboards

Operations [Preview](#)

alarms  0  0  0

logs

[log groups](#) [New](#)

log Anomalies

log Tail

log Insights [New](#)

log Contributor Insights

metrics

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

1m 1h 

UTC timezone ▼

Display ▼



▶	Timestamp	Message
No older events at this moment. Retry		
▶	2025-01-22T16:06:29.477Z	tablename: uber
▶	2025-01-22T16:07:26.072Z	+-----+-----+-----+-----+-----+-----+-----+-----+ Date Time (Local) Eyeball...
▶	2025-01-22T16:07:26.072Z	null 16 11 2 3 4 6 null 17 12 2 3 4 4 null 18 11 1 3 4 7 null 19 13 2 2 3 7 ...
▶	2025-01-22T16:07:26.072Z	wing top 20 rows
▶	2025-01-22T16:07:26.629Z	Running autoDebugger shutdown hook.
No newer events at this moment. Auto retry paused. Resume		